

Synthesis, characterization and ethylene (co-)polymerization behavior of half-titanocene 2-(1-(arylimino)ethyl)quinolin-8-olate chlorides†

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The series of half-titanocene 2-(1-(arylimino)ethyl)quinolin-8-olate dichlorides, Cp'TiCl₂L (Cp' = η^5 -C₅H₅ or η^5 -C₅Me₅, L = 2-(1-(2,6-R¹-4-R²-phenylimino)ethyl)quinolin-8-olate), was synthesized *via* the stoichiometric reaction of Cp'TiCl₃ with the corresponding potassium salt, *viz.* 2-(1-(2,6-R¹-4-R²-phenylimino)ethyl)quinolin-8-olate. All compounds were characterized by elemental analysis, ¹H and ¹³C NMR spectroscopy. The molecular structures of complexes **C2** and **C4** were determined by single-crystal X-ray diffraction. When activated with methylaluminoxane (MAO), all half-titanocene pre-catalysts exhibited high activities for ethylene polymerization and co-polymerization with α -olefins or norbornene.

1. Introduction

Olefin-based polymers, such as polyethylenes (PEs), polypropylenes (PPs), and ethylene/ α -olefin co-polymers are the most important to our daily life and thus are by far the most currently produced synthetic polymers. It is widely believed that the design of efficient transition metal catalysts, that precisely control olefin coordination polymerization, can play an important role in the success of this effort, and recent progress of newly designed catalysts offers new possibilities. Since the 1990s, more attention has been devoted to developing pre-catalysts of early-transition¹ and late-transition metal complexes² for olefin polymerization and co-polymerization. Bridged half-metallocene pre-catalysts such as the well-known constrained-geometry complexes (CGC) represent good examples of single-site catalysts, and produce polyolefins with narrow molecular weight distributions as well as exhibiting good incorporation of co-monomers.³ In extending the synthesis of half-metallocene type pre-catalysts, a series of non-bridged half-metallocene pre-catalysts, Cp'M(L)X_n (M = Ti, Zr, Hf; L = anionic ligand; X = halogen or alkyl), have been developed by the Nomura group.⁴ Other anionic

ligands have also been explored including pyridinylalkyloxy,⁵ acetamidinato,⁶ aryloxy,⁷ iminophenoxy,⁸ ketimide,⁹ and N-substituted 2-(iminomethyl)pyrrolides.¹⁰ Beyond half-metallocene pre-catalysts, the bis(phenoxyimino)titanium complexes reported by the Fujita and Coates groups have been shown to exhibit high activity for the polymerization of both ethylene and propylene.¹¹ Recently, a series of 2-(1-(arylimino)propyl)quinolin-8-ols were successfully prepared in our group, their aluminium complexes showed high activity in the ring-opening polymerization of ϵ -caprolactone,¹² whilst the trichloro-titanium complexes thereof exhibited high activities in ethylene homo-polymerization and co-polymerization.^{13a} Half-titanocene dichloride complexes bearing 2-(1-(arylimino)propyl)quinolin-8-olates also showed high activities towards ethylene polymerization and co-polymerization,^{13b} but they exhibited severe thermo-stability problems at elevated temperatures, which is problematic with regard to industrial considerations. Herein, a series of 2-(1-(arylimino)ethyl)quinolin-8-ols are synthesized and used in preparing the half-titanocene 2-(1-(arylimino)ethyl)quinolin-8-olate chlorides. The newly synthesized half-titanocene pre-catalysts are found to also perform well in the polymerization of ethylene and co-polymerization of ethylene with α -olefins or norbornene, but importantly at higher reaction temperatures.

2. Results and discussion

2.1 Synthesis and characterization of half-titanocene complexes (C1–C8)

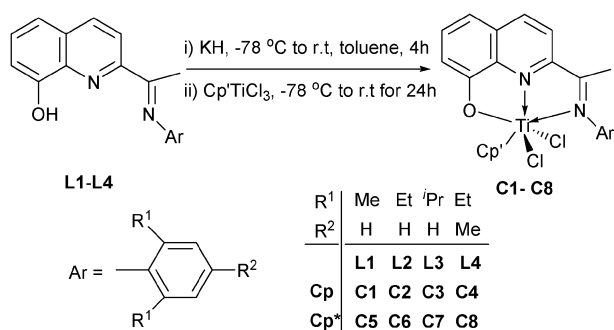
As for the synthetic procedure used for the 2-(1-(arylimino)propyl)quinolin-8-ol,¹² the 2-(1-(arylimino)ethyl)quinolin-8-ols were prepared in good yields *via* the reaction of 2-acetylquinolin-8-ol with the corresponding anilines.¹³ As before, a

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Scheme 1 Synthesis of complexes **C1–C8**.

molar equivalent of potassium hydride and the corresponding 2-(1-(arylimino)ethyl)quinolin-8-ol were mixed in toluene at -78°C and stirred for 4 h, then one equivalent of $\text{Cp}'\text{TiCl}_3$ ($\text{Cp}' = \text{Cp}$ or Cp^*) was added at -78°C and the system was allowed to warm up to room temperature and was stirred for 24 h. The resultant solution was concentrated and extracted with dichloromethane, and the combined filtrate was concentrated and layered with *n*-heptane to precipitate the formed title complex. The title complexes ($\text{Cp}'\text{TiCl}_2$, **C1–C8**) were obtained in good yields (63.1–94.5%) (Scheme 1). All the 2-(1-(arylimino)ethyl)quinolin-8-ols and their half-titanocene dichlorides were characterized by elemental analysis and NMR measurements.

Single crystals of complexes **C2** and **C4** suitable for X-ray crystallographic studies were obtained from dichloromethane solutions when layered with *n*-heptane. The molecular structures are illustrated in Fig. 1 and 2, respectively, and the selected bonds and angles are tabulated in Table 1.

As shown in Fig. 1, the complex **C2** contains one tridentate 2-(1-(2,6-diethylphenylimino)ethyl)quinolin-8-olate, two chlorides and one Cp ring, which form a pseudo octahedral geometry at the titanium center. Due to their different environments, the Ti–Cl and Ti–N bond lengths are slightly varied with Ti1–Cl1 2.4196(15) Å vs. Ti1–Cl2 2.4551(14) Å and Ti1–N1 2.135(4) Å vs. Ti1–N2 2.376(4) Å. The distance between the η^5 -Cp centroid and the titanium is 2.078 Å, which is similar to observations for reported half-titanocene complexes,¹⁴ though there is a slight difference with $\Delta\text{Ti–C}_{\text{Cp}}$ at 0.093 Å with the Ti–C_{Cp} *i.e.* bond lengths within the range

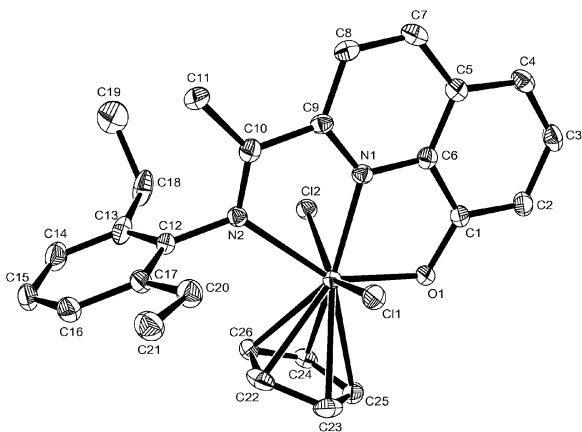


Fig. 1 ORTEP drawing of the molecular structure of **C2** (ellipsoids enclose 30% electronic density; H atoms are omitted for clarity).

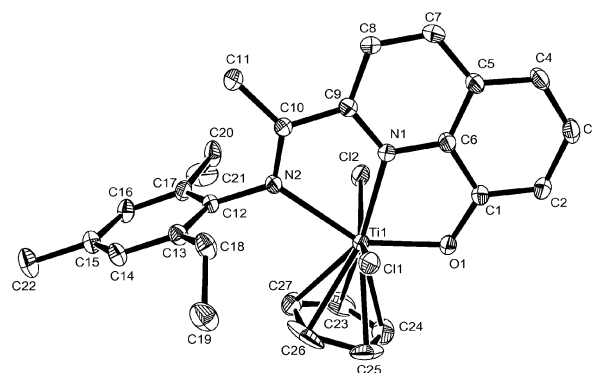


Fig. 2 ORTEP drawing of the molecular structure of **C4** (ellipsoids enclose 30% electronic density; H atoms are omitted for clarity).

Table 1 Selected bond lengths (Å) and angles ($^{\circ}$) for **C2** and **C4**

	C2	C4
Bond lengths (Å)		
Ti1–O1	1.952(3)	1.957(2)
Ti1–N1	2.135(4)	2.133(3)
Ti1–N2	2.376(4)	2.380(3)
Ti1–Cl1	2.4196(15)	2.4367(12)
Ti1–Cl2	2.4551(14)	2.4383(12)
Ti1–Cp _{cent}	2.078	2.071
Bond angles ($^{\circ}$)		
O1–Ti1–N1	77.40(14)	77.19(10)
O1–Ti1–N2	146.37(13)	146.67(10)
N1–Ti1–N2	68.98(14)	69.51(10)
O1–Ti1–Cl1	89.05(10)	88.41(8)
N1–Ti1–Cl1	76.50(11)	76.04(8)
N2–Ti1–Cl1	84.32(10)	82.61(7)
O1–Ti1–Cl2	87.09(10)	89.01(8)
N1–Ti1–Cl2	75.50(11)	75.40(9)
N2–Ti1–Cl2	83.63(9)	83.92(7)
Cl1–Ti1–Cl2	151.91(5)	151.18(4)

of 2.353(5) to 2.446(5) Å. The dihedral angle defined by the two chelating planes [O1–Ti1–Cl1–C6–N1 and O1–Ti1–N1–C9–C10–N2] is only 1.29° , indicating an almost coplanar feature of the atoms of titanium, coordination oxygen and nitrogen atoms as well as the quinolyl ligand framework. The dihedral angle of the phenyl plane of the arylamine and the quinolyl plane is about 86.95° , closer to perpendicular.

The complex **C4** (Fig. 2) having an additional *p*-methyl on the arylamino group possesses similar structural features to **C2** (Fig. 1). Slightly different observations within **C4** are worthy of mention, specifically the near equal Ti–Cl bond lengths of Ti1–Cl1 2.4367(12) Å and Ti1–Cl2 2.4383(12) Å, and the near coplanarity of the titanium and coordinated atoms of the ligands. Furthermore, the quinolyl plane has the dihedral angle of 0.53° formed by the two chelating planes [O1–Ti1–Cl1–C6–N1 and O1–Ti1–N1–C9–C10–N2]. The dihedral angle of the quinolyl and phenyl of arylamine is 88.11° .

2.2 Catalytic behavior in ethylene polymerization

Various alkylaluminium reagents such as MAO, MMAO, EtAlCl_2 and Et_2AlCl have been screened for their suitability as co-catalysts, and based on the observations of the catalytic behavior exhibited by the analogs,¹³ MAO was deemed as the optimum co-catalyst herein. Due to its ease of preparation,

Table 2 Ethylene polymerization by the C7/MAO system^a

Run	Al/Ti	T/°C	t/min	Polym (g)	Act. ^b	M _w ^{c,d}	M _w /M _n ^d	T _m ^e /°C
1	2000	60	30	0.412	4.12	3.94	11.7	127.7
2	3000	60	30	1.293	12.9	4.70	13.1	128.4
3	4000	60	30	1.128	11.3	4.62	14.5	127.7
4	5000	60	30	0.957	9.57	4.26	11.9	130.4
5	3000	60	10	0.173	5.19	3.48	9.49	128.1
6	3000	60	20	0.458	6.87	4.76	13.9	127.8
7	3000	60	60	2.104	10.5	5.24	18.7	128.6
8	3000	20	30	0.998	9.98	5.43	5.11	128.9
9	3000	40	30	1.05	10.5	5.07	6.25	128.2
10	3000	80	30	0.462	4.62	3.28	13.8	126.6

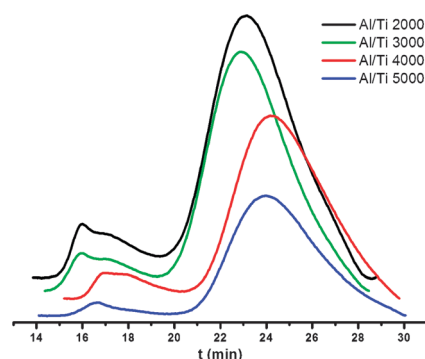
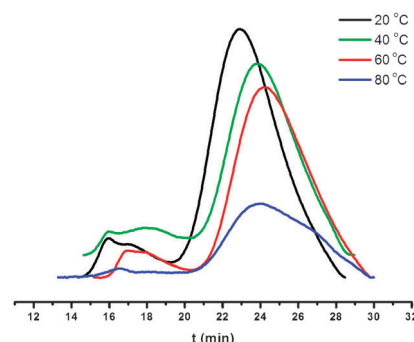
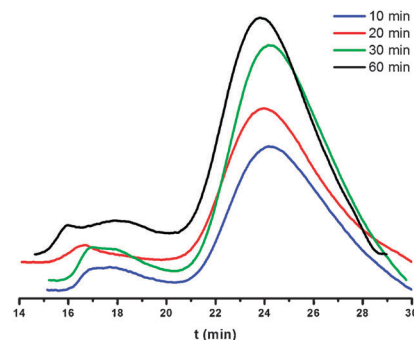
^a Conditions: 2 μmol complex C7, toluene (total volume 100 mL), 10 atm. ^b 10⁵ g mol⁻¹(Ti) h⁻¹. ^c 10⁴ g mol⁻¹. ^d Determined by GPC. ^e Determined by DSC.

complex C7 was employed in the optimization of the ethylene polymerization parameters such as the Al/Ti molar ratio, the reaction temperature and the lifetime of active species (Table 2).

Varying the Al/Ti molar ratio from 2000 to 5000 at 60 °C over 30 min (Runs 1–4, Table 2), the highest activity of 1.29×10^6 g mol⁻¹(Ti) h⁻¹ was achieved at Al/Ti 3000. Regarding the lifetime of the active species, the ethylene polymerization was terminated at different reaction times of 10, 20, 30, and 60 minutes (Runs 2 and 5–7, Table 2). Comparison of these data indicated the presence of an induction period for forming the active species, and thus the activities obtained increased on prolonging the reaction time (Runs 5, 6, and 2, Table 2). In general, the active species in the current system showed longer lifetime than the previously reported analogues.¹³ However, slow deactivation also occurred as observed by the relatively lower values for the catalytic activity over 60 *cf.* 30 min. Considering the thermo-stability of the catalytic system, the temperature parameters were elevated from 20 °C to 80 °C (Runs 2 and 8–10, Table 2), the optimum temperature was found to be 60 °C (Run 2, Table 2). These trends for the reaction parameters were consistent with those observed for their analogs¹³ and other half-titanocene pre-catalysts.¹⁵ The T_m values and molecular weights of the resultant polyethylenes were measured by the DSC and GPC analysis, respectively. In most cases, the broad molecular weight distributions of the polymers indicated the presence of multiply active species in the catalytic system. Generally, the results showed that polyethylenes with higher molecular weights were formed with catalytic systems either at lower reaction temperature or on prolonging the reaction time, whilst the molecular distributions of the polyethylenes increased on either prolonging the reaction time (Fig. 5), increasing the molar ratio of Al/Ti (Fig. 3), or elevating the reaction temperature (Fig. 4).

On comparison with the analog pre-catalysts,^{13b} it can be seen that the current system provided polyethylenes with lower molecular weights.

Using the optimum polymerization conditions of 10 atm ethylene, 60 °C, and Al/Ti 3000 over 30 min, all pre-catalysts C1–C8 were screened for ethylene polymerization (Table 3) in order to further understand the ligand substituent influences on the catalytic activities of these complexes. The most significant influence was observed for the η⁵-Cp group,

**Fig. 3** GPC profiles of PEs obtained from Runs 1–4 in Table 2.**Fig. 4** GPC profiles of PEs obtained from Runs 2, and 8–10 in Table 2.**Fig. 5** GPC profiles of PEs obtained from Runs 2, and 5–7 in Table 2.**Table 3** Ethylene polymerization by the C1–C8/MAO system^a

Run	Cats	Polym (g)	Act. ^b	M _w ^{c,d}	M _w /M _n ^d	T _m ^e /°C
1	C1	0.421	4.21	4.19	12.7	127.6
2	C2	0.637	6.37	n.d. ^f	n.d. ^f	127.7, 141.2
3	C3	0.889	8.89	5.33	11.2	126.9
4	C4	0.610	6.10	5.72	15.3	128.4
5	C5	0.652	6.52	4.85	12.8	127.4
6	C6	1.07	10.7	n.d. ^f	n.d. ^f	129.2, 142.8
7	C7	1.293	12.9	4.70	13.1	130.4
8	C8	0.717	7.17	7.94	20.9	133.2

^a Conditions: 2 μmol complex, Al/Ti = 3000, 60 °C, 30 min, toluene (total volume 100 mL), 10 atm. ^b 10⁵ g mol⁻¹(Ti) h⁻¹. ^c 10⁴ g mol⁻¹. ^d Determined by GPC. ^e Determined by DSC. ^f n.d.: not determined due to partly insoluble polymers.

whereby the pre-catalysts containing $\eta^5\text{-C}_5\text{Me}_5$ (Cp^*) exhibited higher activities than did their analogs bearing $\eta^5\text{-C}_5\text{H}_5$ (Cp), which was thought to be due to the methyl-groups of $\eta^5\text{-Cp}^*$ enhancing the solubility of the half-titanocene complexes, together with the more electron-donating feature stabilizing the active species.⁹ Similar influences of the substituents could also be observed in the catalytic tendencies of these groups of half-titanocene pre-catalysts, in which the pre-catalysts containing Cp^* exhibited the activity order of **C7** (*i*Pr) > **C6** (Et) > **C5** (Me), whilst the pre-catalysts bearing Cp showed the order of **C3** (*i*Pr) > **C2** (Et) > **C1** (Me). The catalytic performance exhibited by the half-titanocene species presented herein is consistent with that observed for their analogs,^{13b} but the current systems importantly show improvement in thermo-stability. Specifically, the systems herein were optimized at 60 °C, whereas for the analogous systems the preferred temperature was 20 °C.^{13b} However, repeated experimental results still indicated a negative effect for the *para*-methyl on the arylimino group with the activities observed as **C2** ($\text{R}^1 = \text{Et}$, $\text{R}^2 = \text{H}$) > **C4** ($\text{R}^1 = \text{Et}$, $\text{R}^2 = \text{Me}$) and **C6** ($\text{R}^1 = \text{Et}$, $\text{R}^2 = \text{H}$) > **C8** ($\text{R}^1 = \text{Et}$, $\text{R}^2 = \text{Me}$). Beyond that, another strange issue arose in that there were two melting points for the polyethylenes produced by **C2** and **C6**, which were partly insoluble in 1,2,4-trichlorobenzene at 135 °C, caused by the presence of the ultrahigh molecular weight polyethylenes with T_m over 140 °C.

2.3 Ethylene/1-hexene and ethylene/1-octene co-polymerization

The co-polymerization of ethylene with 1-hexene or 1-octene was also explored using the representative pre-catalyst **C7** (Table 4). According to the results of co-polymerizing ethylene with 1-octene (Runs 1–4, Table 4), the catalytic activities decreased along with increasing co-monomer concentration, whilst polymers with higher molecular weights and broader molecular distributions were formed. The broader molecular distribution and lower activity indicated that termination partly happened during the insertion of 1-octene. The DSC analysis showed the potential for higher branches in the polymers for which the T_m values were lowered.

As a comparison, the co-polymerization of ethylene with 1-hexene was conducted and showed relatively better activity results (Run 5 vs. Run 3, Table 4). The T_m values of the co-polymers of ethylene with 1-hexene were lower, however, partly insoluble polymers were also observed, and so further investigations were not carried out.

Table 4 Co-polymerization of ethylene with 1-olefin by the **C7**/MAO system^a

Run	Monomer	Polym (g)	Activity ^b	$M_w^{c,d}$	$M_w/M_n^{d,f}$	$T_m^{e/^\circ\text{C}}$
1	0.1 M C_8^-	0.397	3.97	4.44	19.5	124.4
2	0.4 M C_8^-	0.315	3.15	5.17	19.2	123.7
3	0.7 M C_8^-	0.133	1.33	10.0	30.4	121.7
4	1 M C_8^-	0.042	0.42	20.7	58.6	122.2
5	0.7 M C_6^-	0.283	2.83	n.d. ^f	n.d. ^f	121.4

^a Conditions: 2 μmol complex **C7**, Al/Ti = 3000, 60 °C, 30 min, toluene (total volume 100 mL), 10 atm. ^b $10^5 \text{ g mol}^{-1}(\text{Ti}) \text{ h}^{-1}$. ^c 10^4 g mol^{-1} . ^d Determined by GPC. ^e Determined by DSC. ^f n.d.: not determined due to partly insoluble polymers.

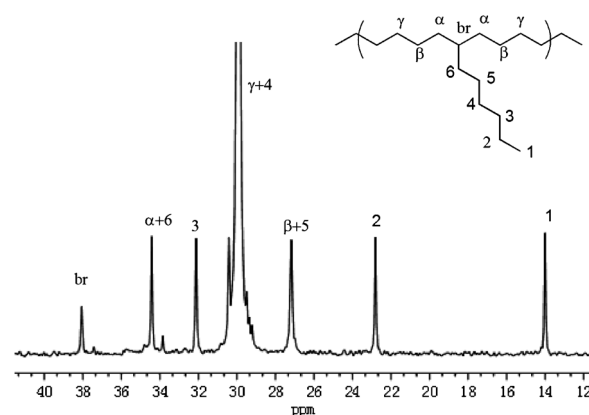


Fig. 6 ^{13}C NMR spectrum of the ethylene/1-octene co-polymer (Run 3, Table 4).

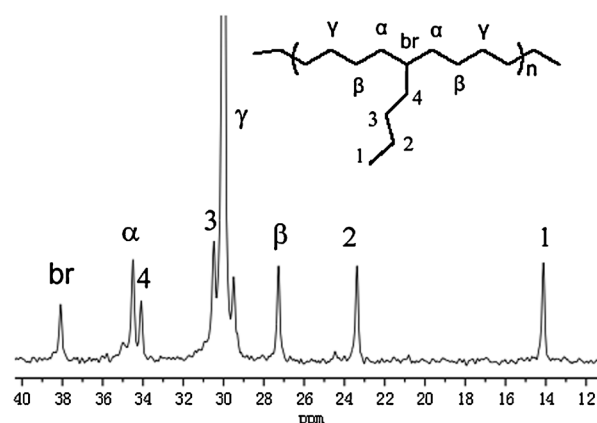


Fig. 7 ^{13}C NMR spectrum of the ethylene/1-hexene co-polymer (Run 5, Table 4).

To determine the degree of incorporation of 1-olefin in the co-polymers, the ^{13}C NMR spectra of the samples of ethylene with 1-octene (Run 3, Table 4) or 1-hexene (Run 5, Table 4) were measured.¹⁶ As shown in Fig. 6, there was 4.22 mol% incorporation of 1-octene, whilst the co-polymer of ethylene/1-hexene contained 10.1 mol% incorporation of 1-hexene (Fig. 7).

Thus, the current catalytic system produced co-polymers with higher incorporation of 1-hexene than 1-octene under the same reaction conditions, but with a lower incorporation of 1-olefins than the observation by the analogous half-titanocene pre-catalysts bearing the 2-(1-(arylimino)propyl)quinolin-8-olate ligand set.^{13b}

2.4 Co-polymerization of ethylene with norbornene

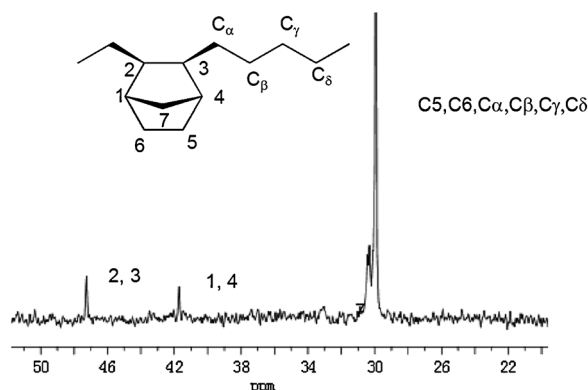
In addition, the co-polymerization of ethylene with norbornene was investigated in order to understand the incorporation ability of cyclic olefins by the **C7**/MAO system; the results are collected in Table 5. As for the observations for the co-polymerization of ethylene with 1-octene, the co-polymerization activities of ethylene with norbornene slowly decreased on increasing the concentration of norbornene, consistent with previous reports in the literature.^{13b} Due to the incorporation of the cyclic monomer, the rigidity of the polymer was increased with slightly different values of T_m , though the

Table 5 Co-polymerization of ethylene with norbornene by the C7/MAO system^a

Run	Monomer	Polym (g)	Activity ^b	$M_w^{c,d}$	M_w/M_n^d	$T_m^{e/°C}$
1	0.1 M	0.348	3.48	9.81	24.3	125.2
2	0.4 M	0.303	3.03	5.49	15.4	124.4
3	0.7 M	0.256	2.56	4.42	18.0	123.7
4	1.0 M	0.251	2.51	n.d. ^f	n.d. ^f	121.4, 144.2

^a Conditions: 2 μ mol Ti, Al/Ti = 3000, 60 °C, 30 min, toluene (total volume 100 mL), 10 atm. ^b 10^5 g mol⁻¹(Ti) h⁻¹. ^c 10^4 g mol⁻¹.

^d Determined by GPC. ^e Determined by DSC. ^f n.d.: not determined due to partly insoluble polymers.

**Fig. 8** ¹³C NMR spectrum of the ethylene/norbornene co-polymer (Run 3, Table 5).

GPC analysis showed that the copolymers possessed lower molecular weights and broader molecular distributions.

According to Fig. 8, the ¹³C NMR spectrum of the ethylene/norbornene co-polymer (Run 3, Table 5) indicated 5.87 mol% incorporation of norbornene.¹⁷

3. Conclusion

The half-titanocene pre-catalysts (C1–C8) were synthesized and fully characterized. All the titanium pre-catalysts performed with high activities towards the polymerization of ethylene when activated with MAO. The pre-catalysts containing Cp* exhibited higher activities than did those pre-catalysts bearing a Cp group. Substituent effects led to the observed catalytic activity order of C3 (*i*Pr) > C2 (Et) > C1 (Me), C7 (*i*Pr) > C6 (Et) > C5 (Me). Appreciable incorporation of 1-olefin or norbornene (a cyclic olefin) was achieved in co-polymerizations by the pre-catalyst C7, suggesting that the systems described herein have potential for use in varied co-polymerization and in the production of advanced co-polymers. Moreover, the title pre-catalysts exhibited enhanced thermo-stability during catalysis and as such are more appropriate for further industrial consideration compared with their ethyl counterparts.^{13b}

4. Experimental section

4.1 General procedures

All manipulations of air and/or moisture-sensitive compounds were conducted under a nitrogen atmosphere in a glovebox or

using standard Schlenk techniques. Methylaluminoxane (MAO, 1.46 M in toluene) was purchased from Albemarle. Other reagents were purchased from Beijing Reagent Chemicals, and potassium hydride (KH) was washed with hexane before use to remove contained mineral oil. Toluene, *n*-hexane, *n*-heptane and norbornene were refluxed over sodium and benzophenone for 4 h, and then distilled and stored under a nitrogen atmosphere. Dichloromethane (CH₂Cl₂), 1-hexene, and 1-octene were refluxed over calcium hydride, and then collected and stored over activated molecular sieves (4 Å), while CDCl₃ was dried over activated 4 Å molecular sieves. IR spectra were recorded on a Perkin Elmer FT-IR 2000 spectrometer using KBr discs in the range of 4000–400 cm⁻¹. Elemental analysis was performed on a Flash EA 1112 microanalyzer. ¹H and ¹³C NMR spectra were recorded on a Bruker DMX 400 MHz instrument at ambient temperature using TMS as an internal standard. DSC traces of polyethylenes were obtained from the second scanning run on a Perkin-Elmer DSC-7 at a heating rate of 10 °C min⁻¹. ¹³C NMR spectra of the polymers were recorded on a Bruker DMX-300 MHz instrument at 110 °C in deuterated 1,2-dichlorobenzene with TMS as an internal standard. The molecular weights and molecular weight distributions of the polymers were determined by gel permeation chromatography (GPC) using a Waters Alliance GPC 2000 instrument equipped with a refractive index (RI) detector and a set of u-Styragel HT columns of 10⁶, 10⁵, 10⁴, and 10³ pore sizes in series. The measurement was performed at 135 °C with 1,2,4-trichlorobenzene as the eluent at a flow rate of 0.95 mL min⁻¹. Narrow-molecular-weight PS samples were used as standards for calibration.

4.2 Synthesis of 2-(1-(arylimino)ethyl)quinolin-8-olate derivatives (L1–L4)

2-(1-((2,6-Dimethylphenyl)imino)ethyl)quinolin-8-ol (L1). A solution of 2,6-dimethylaniline (0.291 g, 2.4 mmol), 1-(8-hydroxyquinolin-2-yl)ethanone (0.374 g, 2 mmol), and a catalytic amount of *p*-toluenesulfonic acid (0.1 g) in toluene was refluxed for 24 h, then the solvent was removed *in vacuo*. The product, 2-(1-((2,6-dimethylphenyl)imino)ethyl)quinolin-8-ol, was purified by column chromatography (silica gel, petroleum ether/ethyl acetate = 20:1), the second part to elute was collected and concentrated to give a yellow green solid in 81% yield (1.50 g). ¹H NMR (CDCl₃, 400 MHz, ppm): δ 8.56 (d, *J* = 8.6 Hz, 1H, quino-*H*), 8.25 (d, *J* = 8.7 Hz, 1H, quino-*H*), 8.17 (s, 1H, OH), 7.53 (t, *J* = 7.9 Hz, 1H, quino-*H*), 7.40 (d, *J* = 8.2 Hz, 1H, quino-*H*), 7.23 (d, *J* = 7.6 Hz, 1H, quino-*H*), 7.10 (d, *J* = 7.5 Hz, 2H, Ar-*H*), 6.97 (t, *J* = 7.5 Hz, 1H, Ar-*H*), 2.32 (s, 3H, N=CCH₃), 2.06 (s, 6H, Ar-CH₃). ¹³C NMR (CDCl₃, 100 MHz, ppm): δ 166.6, 154.0, 152.5, 148.8, 137.1, 136.6, 129.1, 128.9, 128.1, 125.4, 123.3, 119.6, 118.0, 110.5, 18.1, 16.5. FT-IR (KBr, cm⁻¹): 3398 (w), 2947 (m), 1638 (m), 1503 (m), 1462 (m), 1243 (s), 1191 (m), 1029 (m), 1018 (s), 865 (m), 792 (s), 750 (m), 723 (m), 585 (m). mp: 82–83 °C. Anal. calcd for C₁₉H₁₈N₂O: C, 78.59; H, 6.25; N, 9.65%. Found: C, 78.46; H, 6.33; N, 9.54%.

2-(1-((2,6-Diethylphenyl)imino)ethyl)quinolin-8-ol (L2). Using the above procedure, 2,6-diethylaniline was used instead of

2,6-dimethylaniline, the 2-(1-((2,6-diethylphenyl)imino)ethyl)quinolin-8-ol was obtained as a yellow green solid in 62.4% yield (2.1 g). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.55 (d, J = 8.6 Hz, 1H, quino-*H*), 8.23 (d, J = 8.6 Hz, 1H, quino-*H*), 8.17 (s, 1H, OH), 7.50 (t, J = 7.9 Hz, 1H, quino-*H*), 7.38 (d, J = 8.2 Hz, 1H, quino-*H*), 7.22 (d, J = 7.8 Hz, 1H, quino-*H*), 7.13 (d, J = 7.4 Hz, 2H, Ar-*H*), 7.05 (t, J = 7.5 Hz, 1H, Ar-*H*), 2.47–2.35 (m, 4H, Ar- CH_2CH_3), 2.33 (s, 3H, $\text{N}=\text{CCH}_3$), 1.15 (t, J = 7.5 Hz, 6H, Ar- CH_2CH_3). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 166.4, 154.0, 152.6, 147.8, 137.1, 136.6, 131.1, 129.1, 128.9, 126.2, 123.7, 119.6, 118.0, 110.5, 24.8, 16.9, 13.9. FT-IR (KBr, cm^{-1}): 3440 (m), 3053 (w), 2969 (m), 2933 (m), 2862 (m), 1610 (s), 1563 (s), 1506 (s), 1431 (m), 1335 (s), 1309 (m), 1239 (s), 1199 (s), 1163 (s), 1089 (m), 950 (s), 853 (s), 792 (m), 751 (s), 729 (s), 676 (w), 554 (w). mp: 88–89 °C. Anal. calcd for $\text{C}_{21}\text{H}_{22}\text{N}_2\text{O}$: C, 79.21; H, 6.96; N, 8.80%. Found: C, 79.14; H, 7.03; N, 8.68%.

2-(1-((2,6-Diisopropylphenyl)imino)ethyl)quinolin-8-ol (L3).

Using the above procedure, 2,6-diisopropylaniline was used instead of 2,6-diethyl aniline, the 2-(1-((2,6-diisopropylphenyl)imino)ethyl)quinolin-8-ol was obtained as a yellow green solid in 74.2% yield (1.92 g). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.56 (d, J = 8.7 Hz, 1H, quino-*H*), 8.26 (d, J = 8.7 Hz, 1H, quino-*H*), 8.17 (s, 1H, OH), 7.53 (t, J = 7.9 Hz, 1H, quino-*H*), 7.40 (d, J = 8.1 Hz, 1H, quino-*H*), 7.23 (d, J = 7.6 Hz, 1H, quino-*H*), 7.19 (d, J = 7.2 Hz, 2H, Ar-*H*), 7.12 (t, J = 7.5 Hz, 1H, Ar-*H*), 2.83–2.76 (m, 2H, Ar- $\text{CH}(\text{CH}_3)_2$), 2.38 (s, 3H, $\text{N}=\text{CCH}_3$), 1.19 (d, J = 6.9 Hz, 12H, Ar- $\text{CH}(\text{CH}_3)_2$). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 166.5, 154.0, 152.6, 146.5, 137.1, 136.6, 135.7, 129.2, 128.9, 123.9, 123.2, 119.6, 118.0, 110.5, 28.5, 23.3, 23.0, 17.2. FT-IR (KBr, cm^{-1}): 3430 (m), 3065 (m), 2969 (w), 2930 (m), 2868 (s), 1631 (s), 1565 (m), 1505 (m), 1455 (m), 1437 (s), 1333 (m), 1309 (s), 1231 (s), 1190 (s), 1166 (m), 1088 (m), 1062 (m), 959 (m), 871 (s), 785 (m), 760 (w), 749 (m), 721 (m), 662 (w), 552 (w). mp: 153–154 °C. Anal. calcd for $\text{C}_{23}\text{H}_{26}\text{N}_2\text{O}$: C, 79.73; H, 7.56; N, 8.09%. Found: C, 79.65; H, 7.58; N, 7.97%.

2-(1-((2,6-Diethyl-4-methylphenyl)imino)ethyl)quinolin-8-ol (L4).

Using the above procedure, 2,6-diethyl-4-methylaniline was used instead of 2,6-diisopropylaniline, the 2-(1-((2,6-diethyl-4-methylphenyl)imino)ethyl)quinolin-8-ol was obtained as a yellow green solid in 64.7% yield (1.33 g). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.56 (d, J = 8.7 Hz, 1H, quino-*H*), 8.24 (d, J = 8.7 Hz, 1H, quino-*H*), 8.19 (s, 1H, OH), 7.52 (t, J = 7.9 Hz, 1H, quino-*H*), 7.40 (d, J = 8.2 Hz, 1H, quino-*H*), 7.24 (d, J = 7.6 Hz, 1H, quino-*H*), 6.96 (s, 2H, Ar-*H*), 2.45–2.29 (m, 4H, Ar- CH_2CH_3), 2.36 (s, 3H, Ar- CH_3), 2.34 (s, 3H, $\text{N}=\text{CCH}_3$), 1.14 (t, J = 6.9 Hz, 6H, Ar- CH_2CH_3). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 166.4 ($\text{CH}=\text{N}$), 153.4, 152.6, 145.1, 137.0, 136.7, 132.6, 130.8, 129.0, 128.8, 126.7, 120.5, 118.0, 110.5, 24.8, 21.4, 16.9, 13.8. FT-IR (KBr, cm^{-1}): 3442 (O-H) (m), 3041 (m), 2966 (m), 2929 (m), 2845 (m), 1630 ($\text{CH}=\text{N}$) (s), 1577 (m), 1514 (s), 1428 (s), 1357 (m), 1303 (m), 1246 (m), 1190 (m), 1162 (s), 1076 (m), 934 (s), 852 (s), 791 (m), 757 (w), 722 (m), 665 (w), 592 (w). mp: 121 °C. Anal. calcd for $\text{C}_{22}\text{H}_{24}\text{N}_2\text{O}$: C, 79.48; H, 7.28; N, 8.43%. Found: C, 79.35; H, 7.36; N, 8.41%.

4.3 Synthesis and characterization of half-titanocene 2-(1-(arylimino)ethyl)quinolin-8-olate dichlorides (C1–C8)

Dichloro(η^5 -cyclopentadienyl)titanium 2-(1-(2,6-dimethylphenyl imino)ethyl)quinolin-8-olate (C1). To a stirred solution of 2-(1-(2,6-dimethylphenylimino)ethyl)quinolin-8-ol (0.290 g, 1.00 mmol) in dried toluene (30 mL) at -78 °C, KH (0.040 g, 1.00 mmol) was added. The mixture was stirred for an additional 4 h. At -78 °C, $\text{Cp}'\text{TiCl}_3$ (0.220 g, 1.00 mmol, $\text{Cp}' = \text{Cp}$, Cp^*) was added to form the complex. The resultant mixture was allowed to warm to room temperature and stirred for an additional 24 h. The residue, obtained by removing the solvent under vacuum, was extracted with CH_2Cl_2 (3×20 mL) and the combined filtrates were concentrated in vacuum to reduce the volume to 20 mL. Heptane (45 mL) was layered and several days later, dark red crystals were obtained (0.250 g, yield 63.1%). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.48 (d, J = 8.6 Hz, 1H, quino-*H*), 7.91 (d, J = 8.6 Hz, 1H, quino-*H*), 7.78 (t, J = 8.4 Hz, 1H, quino-*H*), 7.46 (d, J = 8.4 Hz, 1H, quino-*H*), 7.20 (s, 3H, quino-*H*), 6.98 (d, J = 7.8 Hz, 1H, Ar-*H*), 6.74 (s, 5H, Cp), 2.39 (s, 6H, Ar- CH_3), 2.31 (s, 3H, $\text{N}=\text{CCH}_3$). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 170.3 ($\text{CH}=\text{N}$), 163.4, 145.0, 141.3, 133.7, 131.3, 129.8, 129.5, 129.2, 128.8, 128.4, 125.5, 120.6, 111.8, 19.7, 17.2, 15.9. Anal. calcd for $\text{C}_{24}\text{H}_{22}\text{Cl}_2\text{N}_2\text{OTi}$: C, 60.91; H, 4.69; N, 5.92%. Found: C, 60.85; H, 4.77; N, 5.85%.

Dichloro(η^5 -cyclopentadienyl)titanium 2-(1-(2,6-diethylphenyl imino)ethyl)quinolin-8-olate (C2). Using the same procedure as for C1, C2 was prepared by using L2 instead of L1 and isolated as brown crystals in 1.04 g (84.2%). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.48 (d, J = 8.6 Hz, 1H, quino-*H*), 7.91 (d, J = 8.6 Hz, 1H, quino-*H*), 7.78 (t, J = 8.0 Hz, 1H, quino-*H*), 7.45 (d, J = 8.2 Hz, 1H, quino-*H*), 7.34 (s, 3H, quino-*H*), 6.97 (d, J = 7.8 Hz, 1H, Ar-*H*), 6.71 (s, 5H, Cp), 2.97–2.88 (m, 2H, Ar- CH_2CH_3), 2.72–2.62 (m, 2H, Ar- CH_2CH_3), 2.34 (s, 3H, $\text{N}=\text{CCH}_3$), 1.21 (t, J = 7.5 Hz, 6H, Ar- CH_2CH_3). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 170.5 ($\text{CH}=\text{N}$), 154.2, 153.6, 141.4, 138.9, 137.7, 131.3, 129.6, 129.3, 128.9, 124.2, 121.8, 118.8, 111.2, 24.9, 20.3, 18.7. Anal. calcd for $\text{C}_{26}\text{H}_{26}\text{Cl}_2\text{N}_2\text{OTi}$: C, 62.30; H, 5.23; N, 5.59%. Found: C, 62.24; H, 5.26; N, 5.51%.

Dichloro(η^5 -cyclopentadienyl)titanium 2-(1-(2,6-diisopropylphenyl imino)ethyl)quinolin-8-olate (C3). Using the same procedure as for C1, C3 was prepared by using L3 instead of L1 and isolated as brown crystals in 1.08 g (94.2%). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.45 (d, J = 8.7 Hz, 1H, quino-*H*), 7.88 (d, J = 8.7 Hz, 1H, quino-*H*), 7.76 (t, J = 7.9 Hz, 1H, quino-*H*), 7.43 (d, J = 8.1 Hz, 1H, quino-*H*), 7.38 (s, 3H, quino-*H*), 6.95 (d, J = 7.5 Hz, 1H, Ar-*H*), 6.76 (s, 5H, Cp), 3.48–3.39 (m, 2H, Ar- $\text{CH}(\text{CH}_3)_2$), 2.40 (s, 3H, $\text{N}=\text{CCH}_3$), 1.45 (d, J = 6.9 Hz, 6H, Ar- $\text{CH}(\text{CH}_3)_2$), 1.02 (d, J = 6.9 Hz, 6H, Ar- $\text{CH}(\text{CH}_3)_2$). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 173.7, 167.3, 147.1, 142.9, 140.4, 138.9, 138.0, 133.1, 130.6, 128.4, 128.0, 125.3, 125.0, 120.7, 116.3, 111.8, 27.9, 25.4, 24.8, 20.7. Anal. calcd for $\text{C}_{28}\text{H}_{30}\text{Cl}_2\text{N}_2\text{OTi}$: C, 63.53; H, 5.71; N, 5.29%. Found: C, 63.47; H, 5.79; N, 5.26%.

Dichloro(η^5 -cyclopentadienyl)titanium 2-(1-(2,6-diethyl-4-methyl-phenylimino)ethyl)quinolin-8-olate (C4). Using the same procedure as for C1, C4 was prepared by using L4 instead of

L1 and was isolated as brown crystals in 1.14 g (94.5%). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.46 (d, J = 8.7 Hz, 1H, quino-*H*), 7.89 (d, J = 8.7 Hz, 1H, quino-*H*), 7.77 (t, J = 7.9 Hz, 1H, quino-*H*), 7.44 (d, J = 8.1 Hz, 1H, quino-*H*), 7.13 (s, 3H, Ar-*H*), 6.97 (d, J = 7.5 Hz, 1H, quino-*H*), 6.72 (s, 5H, Cp), 2.90–2.81 (m, 2H, Ar- CH_2CH_3), 2.70–2.60 (m, 2H, Ar- CH_2CH_3), 2.42 (s, 3H, $\text{N}=\text{CCH}_3$), 2.32 (s, 3H, Ar- CH_3), 1.19 (t, J = 6.9 Hz, 6H, Ar- CH_2CH_3). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 174.1, 167.3, 146.2, 142.9, 138.0, 137.0, 135.0, 133.2, 130.7, 128.4, 127.8, 125.2, 120.5, 116.4, 111.9, 24.6, 21.2, 19.0, 14.8. Anal. calcd for $\text{C}_{27}\text{H}_{28}\text{Cl}_2\text{N}_2\text{OTi}$: C, 62.93; H, 5.48; N, 5.44%. Found: C, 63.04; H, 5.54; N, 5.35%.

Dichloro(η^5 -pentamethylcyclopentadienyl)titanium 2-(1-(2,6-dimethylphenylimino)ethyl)quinolin-8-olate (C5**).** Using the same procedure as for the synthesis of **C1**, **C5** was obtained as a dark red solid in 56.4% yield (0.31 g). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.47 (d, J = 8.4 Hz, 1H, quino-*H*), 8.24 (d, J = 8.6 Hz, 1H, quino-*H*), 7.55 (t, J = 7.7 Hz, 1H, quino-*H*), 7.44 (d, J = 8.3 Hz, 1H, quino-*H*), 7.23 (d, J = 8.1 Hz, 1H, quino-*H*), 7.11 (d, J = 7.4 Hz, 2H, Ar-*H*), 6.98 (t, J = 7.8 Hz, 1H, Ar-*H*), 2.38 (s, 6H, Ar- CH_3), 2.36 (s, 3H, $\text{N}=\text{CCH}_3$), 2.29 (s, 15H, Cp*). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 168.1, 153.1, 152.4, 148.4, 137.0, 136.7, 129.0, 128.8, 128.0, 125.2, 123.2, 120.3, 118.0, 110.4, 23.1, 19.7, 17.2, 15.9, 12.8. Anal. calcd for $\text{C}_{29}\text{H}_{32}\text{Cl}_2\text{N}_2\text{OTi}$: C, 64.10; H, 5.94; N, 5.16%. Found: C, 64.04; H, 6.01; N, 5.12%.

Dichloro(η^5 -pentamethylcyclopentadienyl)titanium 2-(1-(2,6-diethylphenylimino)ethyl)quinolin-8-olate (C6**).** Using the same procedure as for the synthesis of **C2**, **C6** was obtained as a dark red solid in 69% yield (0.36 g). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.49 (d, J = 8.6 Hz, 1H, quino-*H*), 8.29 (d, J = 8.5 Hz, 1H, quino-*H*), 7.55 (t, J = 7.7 Hz, 1H, quino-*H*), 7.42 (d, J = 8.2 Hz, 1H, quino-*H*), 7.25 (d, J = 8.1 Hz, 1H, quino-*H*), 7.17 (d, J = 8.0 Hz, 2H, Ar-*H*), 7.10 (t, J = 7.8 Hz, 1H, Ar-*H*), 2.51–2.43 (m, 2H, Ar- CH_2CH_3), 2.37 (s, 3H, $\text{N}=\text{CCH}_3$), 2.30 (s, 15H, Cp*), 2.17–2.10 (m, 2H, Ar- CH_2CH_3), 1.22 (t, J = 7.5 Hz, 6H, Ar- CH_2CH_3). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 168.5, 153.3, 152.5, 147.5, 137.1, 136.7, 130.9, 129.0, 128.9, 125.9, 123.5, 120.4, 118.0, 110.4, 24.9, 20.3, 18.7, 12.9. Anal. calcd for $\text{C}_{31}\text{H}_{36}\text{Cl}_2\text{N}_2\text{OTi}$: C, 65.16; H, 6.35; N, 4.90%. Found: C, 65.13; H, 6.38; N, 4.89%.

Dichloro(η^5 -pentamethylcyclopentadienyl)titanium 2-(1-(2,6-diisopropylphenylimino)ethyl)quinolin-8-olate (C7**).** Using the same procedure as for the synthesis of **C3**, **C7** was obtained as a dark red solid in 71% yield (0.45 g). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.45 (d, 1H, J = 8.7 Hz, quino-*H*), 8.28 (d, 1H, J = 8.3 Hz, quino-*H*), 7.56 (t, 1H, J = 7.8 Hz, quino-*H*), 7.49 (d, 1H, J = 8.1 Hz, quino-*H*), 7.21 (d, 1H, J = 7.4 Hz, quino-*H*), 7.17 (d, 2H, J = 7.8 Hz, Ar-*H*), 7.03 (t, 1H, J = 7.5 Hz, Ar-*H*), 2.78–2.72 (m, 2H, Ar- $\text{CH}(\text{CH}_3)_2$), 2.42 (s, 3H, $\text{N}=\text{CCH}_3$), 2.28 (s, 15H, Cp*), 1.23 (d, 6H, J = 7.6 Hz, Ar- $\text{CH}(\text{CH}_3)_2$), 1.15 (d, 6H, J = 7.4 Hz, Ar- $\text{CH}(\text{CH}_3)_2$). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 167.3, 153.3, 152.5, 146.2, 137.1, 136.7, 130.8, 129.0, 128.9, 126.7, 123.0, 120.6, 118.0, 110.4, 27.9, 25.4, 24.8, 20.7, 12.8. Anal. calcd for

$\text{C}_{33}\text{H}_{40}\text{Cl}_2\text{N}_2\text{OTi}$: C, 66.12; H, 6.73; N, 4.67%. Found: C, 66.06; H, 6.84; N, 4.59%.

Dichloro(η^5 -pentamethylcyclopentadienyl)titanium 2-(1-(2,6-diethyl-4-methyl-phenylimino)ethyl)quinolin-8-olate (C8**).** Using the same procedure as for the synthesis of **C4**, **C8** was obtained as a dark red solid in 69.2% yield (0.38 g). ^1H NMR (CDCl_3 , 400 MHz, ppm): δ 8.47 (d, J = 8.6 Hz, 1H, quino-*H*), 8.22 (d, J = 8.3 Hz, 1H, quino-*H*), 7.54 (t, J = 7.9 Hz, 1H, quino-*H*), 7.33 (d, J = 8.2 Hz, 1H, quino-*H*), 7.20 (d, J = 7.8 Hz, 1H, quino-*H*), 6.94 (s, 2H, Ar-*H*), 2.48–2.39 (m, 2H, Ar- CH_2CH_3), 2.44 (s, 3H, $\text{N}=\text{CCH}_3$), 2.36 (s, 15H, Cp*), 2.34 (s, 3H, Ar- CH_3), 2.32–2.26 (m, 2H, Ar- CH_2CH_3), 1.12 (t, J = 7.5 Hz, 6H, Ar- CH_2CH_3). ^{13}C NMR (CDCl_3 , 100 MHz, ppm): δ 168.7, 153.5, 152.5, 145.0, 137.1, 136.7, 132.6, 130.8, 129.0, 128.8, 126.7, 120.5, 118.0, 110.4, 24.6, 21.2, 19.0, 14.8, 12.8. Anal. calcd for $\text{C}_{32}\text{H}_{38}\text{Cl}_2\text{N}_2\text{OTi}$: C, 65.65; H, 6.54; N, 4.79%. Found: C, 65.62; H, 6.59; N, 4.71%.

4.4 Procedures for ethylene polymerization and co-polymerization

A 0.25 L stainless steel autoclave equipped with a mechanical stirrer and a temperature controller was heated in a vacuum at 80 °C and recharged with ethylene three times, then cooled to room temperature under an ethylene atmosphere. A toluene solution of the pre-catalyst (with co-monomer) was transferred into the reactor. After the desired reaction temperature was reached, the required amount of the co-catalyst was added with maintaining the total volume at 100 mL, and then the autoclave was immediately pressurized to 10 atm. The ethylene pressure was kept constant throughout the feeding of ethylene during the reaction. After a set period, the feeding of the ethylene was stopped, and then the autoclave was placed in a water-ice bath for 1 h. The resultant mixture was poured into 10% HCl-ethanol solution, and the polymer formed was collected and washed with water and ethanol several times, and then dried under vacuum to a constant weight.

4.5 X-Ray structure determinations

Crystals of **C2** and **C4** suitable for single-crystal X-ray analysis were obtained by laying *n*-heptane on the dichloromethane solutions. Single-crystal X-ray diffraction for **C2** and **C4** was performed on a Rigaku RAXIS Rapid IP diffractometer with graphite-monochromated Mo K α radiation (λ = 0.71073 Å) at 173(2) K. Cell parameters were obtained by global refinement of the positions of all collected reflections. Intensities were corrected for Lorentz and polarization effects and empirical absorption. The structures were solved by direct methods and refined by full-matrix least-squares on F^2 . All non-hydrogen atoms were refined anisotropically. Structure solution and refinement were performed by using the SHELXL-97 package.¹⁸ Crystal data collection and refinement details are given in Table 6.

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Table 6 Crystallographic data and refinement details for complexes **C2** and **C4**

	C2	C4
Empirical formula	C ₂₆ H ₂₆ Cl ₂ N ₂ O ₂ Ti	C ₂₇ H ₂₈ Cl ₂ N ₂ O ₂ Ti
Formula weight	501.29	515.31
Crystal color	Red	Red
Temperature/K	173(2)	173(2)
Wavelength/Å	0.71073	0.71073
Crystal system	Monoclinic	Monoclinic
Space group	C2/c	P2(1)/c
a/Å	23.852(5)	17.098(3)
b/Å	13.832(3)	8.1446(16)
c/Å	16.890(3)	17.624(4)
α/°	90	90
β/°	118.68(3)	98.70(3)
γ/°	90	90
Volume/Å ³	4888.6(17)	2426.0(8)
Z	8	4
D _{calc} /mg m ⁻³	1.362	1.411
μ/mm ⁻¹	0.590	0.596
F(000)	2080	1072
Crystal size/mm	0.20 × 0.15 × 0.09	0.20 × 0.13 × 0.09
θ Range/°	1.76–25.50	1.20–27.47
Limiting indices	–22 ≤ h ≤ 28, –16 ≤ k ≤ 16, –20 ≤ l ≤ 19	–17 ≤ h ≤ 22, –10 ≤ k ≤ 7, –22 ≤ l ≤ 22
No. of rflns collected	17075	19063
No. of unique rflns	4502	5545
R _{int}	0.0486	0.0485
Completeness to θ (%)	98.8(θ = 25.50)	99.6(θ = 27.47)
Goodness-of-fit on F ²	1.191	1.241
Final R indices [I > 2σ(I)]	R1 = 0.0673, wR2 = 0.1805	R1 = 0.0730, wR2 = 0.1774
R indices (all data)	R1 = 0.0780, wR2 = 0.1862	R1 = 0.0789, wR2 = 0.1850

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